Author Attribution Through Word Frequency Analysis

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Question & Approach

Question: Can word frequencies identify authors?

1. Do works by the same author share word patterns?
2. Can we distinguish authors by word frequencies?
3. Are there preferred words unique to each author?

Approach: 6 novels from Project Gutenberg (Dickens, Austen, Twain, Fitzgerald, Melville)

Problems Encountered

Problem	Solution
HTML tags counted as words	BeautifulSoup extraction
Unicode crashes	errors='replace'
Punctuation attached ("said,")	Regex cleaning
Stop words dominating	NLTK stopword removal

Results

1. SAME AUTHOR (Dickens):

   - Similar vocabulary patterns observed across both works

   - Words like 'said', 'would', 'could' appear frequently in both

   - Differences reflect different story settings and themes

2. DIFFERENT AUTHORS (Austen vs Twain):

   - Austen: More formal, social vocabulary (e.g., 'daughter', 'sister', 'family')

   - Twain: More colloquial, dialect-driven vocabulary (e.g., 'dey', 'gwine')

   - Clear stylistic differences in word choice

3. DIFFERENT AUTHORS (Fitzgerald vs Melville):

   - Fitzgerald: More modern, descriptive vocabulary (e.g., 'eyes', 'face', 'house')

   - Melville: More nautical, philosophical vocabulary (e.g., 'whale', 'sea', 'ship')

   - Period and subject matter strongly influence word choice

Conclusions

Authors have distinctive word "fingerprints"

Same author = consistent style

Different authors = clear vocabulary preferences